

Addition and subtraction of algebraic fractions

One denominator, then the numerators — and the bracket that must be written when subtracting.

LESSON 116–117

2 × 40 MIN

MATEMATIKA 7, §52

S8 7.3

LESSON CLOCK

10'

25'

25'

20'

5'

The same denominator	10 min
Different denominators	25 min
The bracket when subtracting	25 min
Finishing the answer	20 min
Homework	5 min

LEARNING OBJECTIVES

- Add and subtract fractions with the same denominator.
- Reduce to a common denominator first when the denominators differ.
- Bracket the whole numerator that is being subtracted.
- Simplify the answer and state the excluded values.

TEXTBOOK

National	pp. 342–349
Cambridge	Stage 8, pp. 70–74
Workbook	Exercise 7.3

01

Explanation

The same denominator

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}$$

The denominator is written once and the numerators are combined — exactly as with ordinary fractions.

SUM	COMBINED	SIMPLIFIED
$\frac{2x}{5} + \frac{3x}{5}$	$\frac{5x}{5}$	x
$\frac{3}{x} + \frac{4}{x}$	$\frac{7}{x}$	$\frac{7}{x}$
$\frac{x}{x+1} + \frac{1}{x+1}$	$\frac{x+1}{x+1}$	1

ALWAYS LOOK AT THE ANSWER AGAIN

The third line collapsed to 1 only because the numerator turned out to be the denominator. Simplifying the result is part of the question, not an extra.

Different denominators

Reduce to the lowest common denominator first — the whole of the previous lesson — and only then combine the numerators.

STEP	WORKING
the sum	$\frac{1}{x-3} + \frac{2}{x+3}$
the LCD	$x^2 - 9$
rewrite both	$\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$
combine	$\frac{3x-3}{x^2-9}$
factorise and check	$\frac{3(x-1)}{(x-3)(x+3)}$

NOTHING CANCELS HERE – AND THAT IS AN ANSWER TOO

$3(x-1)$ shares no factor with $(x-3)(x+3)$, so the fraction is already in its simplest form. Checking is quick; assuming is what costs marks.

The bracket when subtracting

When a whole numerator is subtracted, it goes inside a bracket, and every sign in it changes.

$$\frac{5x-1}{x+2} - \frac{3x+4}{x+2} = \frac{(5x-1) - (3x+4)}{x+2} = \frac{2x-5}{x+2}$$

WRITTEN	RIGHT?
$(5x - 1) - (3x + 4) = 2x - 5$	yes
$5x - 1 - 3x + 4 = 2x + 3$	no — the +4 was not negated

THE MINUS APPLIES TO THE WHOLE NUMERATOR

This single missing bracket accounts for more lost marks in the algebraic-fractions paper than every other slip combined. Write the bracket first and remove it afterwards.

Finishing the answer

STEP	WHAT TO DO
1	factorise the denominators and find the LCD
2	multiply each fraction up by its extra factor
3	combine the numerators, bracketing anything subtracted
4	collect like terms in the numerator
5	factorise the numerator and cancel if possible
6	state the excluded values

EXPRESSION	ANSWER
$\frac{1}{x} + \frac{1}{x+1}$	$\frac{2x+1}{x(x+1)}$
$\frac{1}{x-1} - \frac{1}{x+1}$	$\frac{2}{x^2-1}$
$\frac{x}{x-2} - \frac{2}{x-2}$	1

02 Worked examples

EXAMPLE 1 Simplify $\frac{1}{x-3} + \frac{2}{x+3}$.

① LCD = $x^2 - 9$, with $x \neq \pm 3$.

② $\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$

③ = $\frac{3x-3}{x^2-9}$

④ = $\frac{3(x-1)}{(x-3)(x+3)}$

Nothing cancels.

ANSWER

$$\frac{3x-3}{x^2-9}$$

EXAMPLE 2 Simplify $\frac{5x - 1}{x + 2} - \frac{3x + 4}{x + 2}$.

① The denominators already match.

② $= \frac{(5x - 1) - (3x + 4)}{x + 2}$

The bracket is compulsory.

③ $= \frac{5x - 1 - 3x - 4}{x + 2}$

Both signs changed.

④ $= \frac{2x - 5}{x + 2}$

ANSWER

$$\frac{2x - 5}{x + 2}$$

EXAMPLE 3 Simplify $\frac{1}{x - 1} - \frac{1}{x + 1}$.

① $\text{LCD} = (x - 1)(x + 1) = x^2 - 1$.

② $= \frac{(x + 1) - (x - 1)}{x^2 - 1}$

③ $= \frac{x + 1 - x + 1}{x^2 - 1}$

④ $= \frac{2}{x^2 - 1}$

ANSWER

$$\frac{2}{x^2 - 1}$$

03 Interactive model

Mark one exercise for the bracket alone, ignoring everything else; the class discovers where its marks actually go.

Adding and subtracting, step by step

$$\frac{1}{x-3} + \frac{2}{x+3}$$

① LCD

$= (x - 3)(x + 3)$
 $= (x^2 - 9)$

② $\frac{x+3}{x^2-9} + \frac{2x-6}{x^2-9}$

Each
fraction
multiplied
up.

③ $\frac{3x-3}{x^2-9}$

Collect
the
numerators.

④ $\frac{3(x-1)}{(x-3)(x+3)}$

Factorise
and
check
for a
cancellation
—
there
is
none.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Addition	Qo'shish	Сложение
Subtraction	Ayirish	Вычитание
Common denominator	Umumiy maxraj	Общий знаменатель
Numerator	Surat	Числитель
Bracket	Qavs	Скобка
Like terms	O'xshash hadlar	Подобные члены
To simplify	Soddalashtirish	Упростить
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

With the same denominator you combine:

the denominators

the numerators

both

neither

$\frac{3}{x} + \frac{4}{x}$ equals:

$$\frac{7}{x}$$

$$\frac{7}{2x}$$

$$\frac{12}{x}$$

$$\frac{7}{x^2}$$

When subtracting, the second numerator needs:

a bracket

a denominator

nothing

a factor

$(5x - 1) - (3x + 4)$ equals:

$\frac{1}{x-1} - \frac{1}{x+1}$ equals:

\frac{2}{x^2 - 1}"/>

\frac{2x}{x^2 - 1}"/>

\frac{1}{x^2 - 1}"/>

The last step of every such question is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.
$$\frac{2x}{5} + \frac{3x}{5}$$

ANSWER x

2.
$$\frac{3}{x} + \frac{4}{x}$$

ANSWER $\frac{7}{x}$

3.
$$\frac{x}{x+1} + \frac{1}{x+1}$$

ANSWER 1

4.
$$\frac{5}{a} - \frac{2}{a}$$

ANSWER $\frac{3}{a}$

5.
$$\frac{x}{x-2} - \frac{2}{x-2}$$

ANSWER 1

6.
$$\frac{7x}{3} - \frac{x}{3}$$

ANSWER $2x$

7.
$$(5x - 1) - (3x + 4)$$

ANSWER $2x - 5$

Medium · the standard question

7 PROBLEMS

1.
$$\frac{1}{x-3} + \frac{2}{x+3}$$

ANSWER
$$\frac{3x-3}{x^2-9}$$

2.
$$\frac{1}{x-1} - \frac{1}{x+1}$$

ANSWER
$$\frac{2}{x^2-1}$$

3.
$$\frac{1}{x} + \frac{1}{x+1}$$

ANSWER
$$\frac{2x+1}{x(x+1)}$$

4.
$$\frac{5x-1}{x+2} - \frac{3x+4}{x+2}$$

ANSWER
$$\frac{2x-5}{x+2}$$

5.
$$\frac{2}{3x} + \frac{1}{2x}$$

ANSWER
$$\frac{7}{6x}$$

6.
$$\frac{1}{x} - \frac{1}{x+2}$$

ANSWER
$$\frac{2}{x(x+2)}$$

7.
$$\frac{3}{x-1} + \frac{3}{1-x}$$

ANSWER
$$0$$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{x-2} + \frac{4}{x^2-4}$

ANSWER $\frac{x+6}{x^2-4}$

2. $\frac{x}{x+1} - \frac{1}{x-1}$

ANSWER $\frac{x^2-2x-1}{x^2-1}$

3. $\frac{1}{x} + \frac{1}{x+1} + \frac{1}{x+2}$ as one fraction

ANSWER $\frac{3x^2+6x+2}{x(x+1)(x+2)}$

4. $\frac{2}{x-3} - \frac{2}{x+3}$

ANSWER $\frac{12}{x^2-9}$

5. $\frac{a}{a-b} + \frac{b}{b-a}$

ANSWER 1

6. $\frac{1}{x^2-1} + \frac{1}{x^2+2x+1}$

ANSWER $\frac{2x}{(x-1)(x+1)^2}$

7. Why is a bracket needed when subtracting?

ANSWER The minus applies to every term of that numerator

07 Homework

Homework — 5 tasks

Write the bracket before you remove it, every single time.

1. Simplify $\frac{2}{x} + \frac{3}{x}$ and $\frac{4x}{7} - \frac{x}{7}$.
2. Simplify $\frac{1}{x-5} + \frac{2}{x+5}$.
3. Simplify $\frac{4x+3}{x-1} - \frac{2x+7}{x-1}$.
4. Simplify $\frac{1}{x} - \frac{1}{x+3}$.
5. Simplify $\frac{5}{x-2} + \frac{5}{2-x}$ and explain the answer.